



[HINT SOLUTION]

Q.1 $\frac{2}{9} \sin 20^\circ$

Sol.

$$S = w + 2w^2 + 3w^3 + \dots + 9w^9$$

$$Sw = w^2 + 2w^3 + \dots + 8w^9 + 9w^{10} \text{ where } w^9 = (\cos 40^\circ + i \sin 40^\circ)^9 = 1 \text{ and } |w| = 1$$

$$S(1 - w) = w + w^2 + w^3 + \dots + w^9 - 9w^{10}$$

$$= \frac{w(1 - w^9)}{1 - w} - 9w = 0 - 9w$$

$$S = -\frac{9w}{1 - w} \text{ (using } w^9 = 1); \left| \frac{1}{S} \right| = \left| \frac{w - 1}{9w} \right| = \frac{1}{9} |\cos 40^\circ + i \sin 40^\circ - 1|$$

$$= \frac{1}{9} |-2 \sin^2 20^\circ + 2 + i \sin 20^\circ \cos 20^\circ| = \frac{1}{9} |2 \sin^2 20^\circ i (\cos 20^\circ + i \sin 20^\circ)| = \frac{2}{9} \sin 20^\circ \text{ Ans.}$$

Q.2 54

Sol. Answer is 54

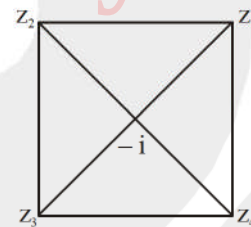
Q.3 $2^{5/4}$

Sol. $z^4 + 4z^3i + 6z^2i^2 + 4zi^3 + i^4 = 1 + i$

$$(z + i)^4 = 1 + i \Rightarrow |z + i|^4 = \sqrt{2} \Rightarrow |z + i| = 2^{1/8}$$

$$\text{Area} = \frac{d^2}{2} = \frac{4|z_1 + i|^2}{2}$$

$$= 2 \cdot 2^{1/8} \cdot 2^{1/8} = 2^{5/4} \text{ Ans.}$$



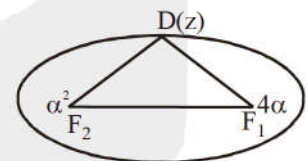
Q.4 3

Sol. We must have $|\alpha^2 - 4\alpha| < 5 \Rightarrow -5 < \alpha^2 - 4\alpha < 5$

$$\Rightarrow -1 < (\alpha - 2)^2 < 9 \Rightarrow 0 \leq (\alpha - 2)^2 < 9 \Rightarrow -3 < \alpha - 2 < 3$$

$$\Rightarrow -1 < \alpha < 5 \Rightarrow 0 < \alpha < 5 \Rightarrow \alpha = 1, 2, 3, 4 \Rightarrow 4 \text{ values.]}$$

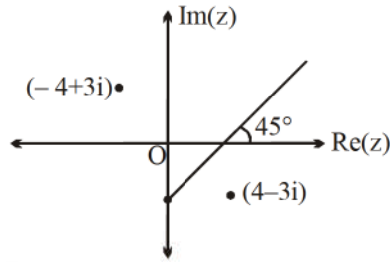
[Note : $\alpha = 0, 4$ is not possible because, $|z - \alpha^2| + |z - 4\alpha| = 5$ will represent a circle.]



Q.5 10

Sol. As, z lies on the curve $\arg(z + i) = \frac{\pi}{4}$, which is a ray originating from $(-i)$ and lying right side of imaginary axis making an angle $\frac{\pi}{4}$ with the real axis in anticlockwise sense.





∴ The value of $|z - (-4 + 3i)| + |z - (4 - 3i)|$ will be minimum when $z, -4 + 3i, 4 - 3i$ are collinear.
 ∴ Minimum value = distance between $(-4 + 3i)$ and $(4 - 3i)$

$$= \sqrt{(-4-4)^2 + (3+3)^2} = \sqrt{64+36} = \sqrt{100} = 10.$$

Q.6 $\alpha = -1$ and $\beta = 1$

Sol. $x^2 - x + 1 = 0 \Rightarrow x_1 = -w$ and $x_2 = -w^2$ where w is the cube root of unity

$$\therefore \alpha = w^{2008} + w^{4016} = w + w^2 = -1$$

$$\beta = -[w^{2009} + w^{4018}] = -[w^2 + w] = 1$$

hence $\alpha = -1$ and $\beta = 1$ **Ans.**

Q.7 0

Sol.

$$\begin{aligned} Z &= \cos 12^\circ + \cos 48^\circ + i(\sin 48^\circ + \sin 12^\circ) \\ &= 2 \cos 30^\circ \cdot \cos 18^\circ + 2i \sin 30^\circ \cdot \cos 18^\circ \\ &= 2 \cos 18^\circ (\cos 30^\circ + i \sin 30^\circ) \\ &= 2 \cos 18^\circ [\cos(\pi/6) + i \sin(\pi/6)] \end{aligned}$$

$$\begin{aligned} Z^6 &= 2^6 \cdot \cos^6 18^\circ (\cos \pi + i \sin \pi) \\ &= -2^6 (\cos^6 18^\circ) + 0i \end{aligned}$$

$$\text{Im } Z^6 = 0 \text{ **Ans.**}$$

Q.8 If $\arg(z+3) = \frac{\pi}{6}$ and $\arg(z-3) = \frac{2\pi}{3}$, then $\tan^2(\arg z) - 2 \cos(\arg z)$, is

(A) 0

(B) 1

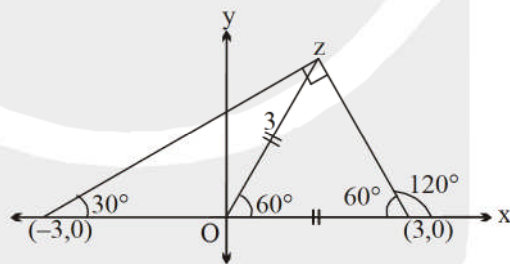
(C*) 2

(D) 3

Sol.

$$\text{From the figure } z = 3e^{i\frac{\pi}{3}}$$

$$\therefore \arg z = \frac{\pi}{3}$$



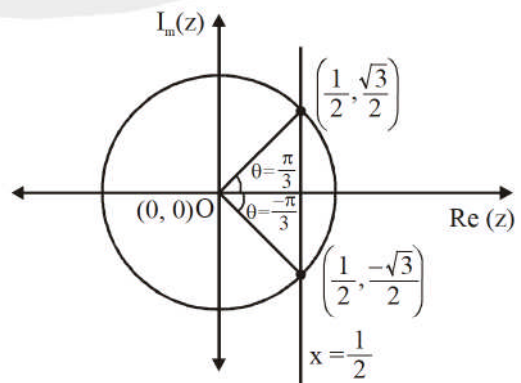
$$\text{Hence } \tan^2(\arg z) - 2 \cos(\arg z) = \tan^2 60^\circ - 2 \cos 60^\circ = 3 - 1 = 2 \text{ **Ans.**}$$

Q.9 $\frac{\pi}{3}$

Sol. $|z| = 1 \Rightarrow z = x + iy$ lies on circle $x^2 + y^2 = 1$.

$$\text{Also, } z + \bar{z} = 1 \Rightarrow x = \frac{1}{2}$$

$$\therefore z = \frac{1+i\sqrt{3}}{2} \text{ or } \frac{1-i\sqrt{3}}{2}$$



$\Rightarrow$ Principal argument of z can be $\frac{\pi}{3}$ or $\frac{-\pi}{3}$.

Q.10 $\sqrt{5}$

Sol. Put $z = x + iy$, in the given equation and then on comparing real and imaginary parts, we get

$$z = -2 - i \Rightarrow |z| = \sqrt{5}$$

Q.11 (A) $|z| + 2|\omega| = 10$

(D) $2|z|^2 - |\omega|^2 = 23$

Sol. Let $|z|^2 = x$, $|\omega|^2 = y$ and $z\bar{\omega} + \bar{z}\omega = t$.

$$|2z - \omega| = 25 \Rightarrow (2z - \omega)(2\bar{z} - \bar{\omega}) = 25$$

$$4x + y - 2t = 25 \quad \dots\dots(i)$$

$$|z + 2\omega| = 10 \Rightarrow x + 4y + 2t = 100 \quad \dots\dots(ii)$$

$$|z + \omega| = 7 \Rightarrow x + y + t = 49 \quad \dots\dots(iii)$$

Add (i) and (ii) to get $x + y = 25$

Hence, $t = 24$, $x = 16$, $y = 9$

$\therefore |z| = 4$ and $|\omega| = 3$

Q.12 (B) $\alpha\bar{\beta} = \beta\bar{\alpha}$

(C) $\alpha\bar{\beta} = 1$

Sol. $|\alpha|^2\beta - |\beta|^2\alpha = \alpha - \beta \quad \dots(1)$

$$\beta(1 + |\alpha|^2) = \alpha(1 + |\beta|^2)$$

$$\frac{\alpha}{\beta} = \frac{1 + |\alpha|^2}{1 + |\beta|^2} \Rightarrow \frac{\alpha}{\beta} \text{ is positive real} \Rightarrow \boxed{\text{amp}\left(\frac{\alpha}{\beta}\right) = 0}$$

$$\text{Also } \frac{\alpha}{\beta} = \frac{\bar{\alpha}}{\bar{\beta}} \Rightarrow \boxed{\alpha\bar{\beta} = \bar{\alpha}\beta}$$

$$\text{again from (1) } \alpha\bar{\alpha}\beta - \beta\bar{\beta}\alpha = \alpha - \beta$$

$$\alpha(\bar{\alpha}\beta - 1) = \beta(\alpha\bar{\beta} - 1)$$

but $\bar{\alpha}\beta = \alpha\bar{\beta}$, (As α and β are distinct complex numbers)

$$\text{hence } (\bar{\alpha}\beta - 1)(\alpha - \beta) = 0$$

$$\alpha \neq \beta \Rightarrow \bar{\alpha}\beta = 1 = \alpha\bar{\beta} \Rightarrow \text{(B) and (C) are correct]$$

Q.13 (A) maximum value of $|z_1 + iz_2| = 17$.

(B) minimum value of $|z_1 + (1 + i)z_2| = 13 - 4\sqrt{2}$.

$$(C) \text{ maximum value of } \left| \frac{z_1}{z_2 + \frac{4}{z_2}} \right| = \frac{13}{3}.$$

Sol. $z_1 = 5 + 12i$, $|z_2| = 4$

$$|z_1 + iz_2| \leq |z_1| + |z_2| = 13 + 4 = 17$$

$$\therefore |z_1 + (1+i)z_2| \geq ||z_1| - |1+i||z_2|| = 13 - 4\sqrt{2}$$

$$\therefore \min |z_1 + (1+i)z_2| = 13 - 4\sqrt{2}$$

$$\text{Also, } \left| z_2 + \frac{4}{z_2} \right| \leq |z_2| + \frac{4}{|z_2|} = 4 + 1 = 5$$

$$\text{and } \left| z_2 + \frac{4}{z_2} \right| \geq \left| |z_2| - \frac{4}{|z_2|} \right| = 4 - 1 = 3$$

$$\Rightarrow \max \left| \frac{z_1}{z_2 + \frac{4}{z_2}} \right| = \frac{13}{3} \quad \text{and} \quad \min \left| \frac{z_1}{z_2 + \frac{4}{z_2}} \right| = \frac{13}{5} \quad \text{Ans.}$$

Q.14 (B) $z_1, z_2, z_3, \dots, z_n$ are collinear points

$$(C) \sum_{r=1}^n \operatorname{Re}(z_r) = \frac{-7}{2}$$

$$(D) \sum_{r=1}^n \operatorname{Im}(z_r) = 0$$

Sol. We have $|z|(z+1)^8 = z^8|z+1| \dots (1)$

Taking modulus on both sides, we get

$$|z||z+1|^8 = |z|^8|z+1|$$

$$\therefore |z+1|^7 = |z|^7$$

$$\Rightarrow |z+1| = |z|$$

which represents the locus of z will be a straight line which is perpendicular bisector of the line segment joining $(-1, 0)$ and $(0, 0)$ i.e. $\operatorname{Re}(z) = \frac{-1}{2}$. Also there will be exactly 7 distinct 'z' satisfying given equation,

$$\text{i.e. } z = \frac{-1}{2}, \frac{-1}{2} \pm ki \text{ where } k \text{ has 3 distinct positive values } k_1, k_2 \text{ and } k_3.$$

$$\therefore \sum_{r=1}^n \operatorname{Re}(z_r) = \sum_{r=1}^7 z_r = \frac{-1}{2} \times 7 = -\frac{7}{2}$$

$$\text{and } \sum_{r=1}^n \operatorname{Im}(z_r) = 0 + (k_1 + k_2 + k_3) + (-k_1 - k_2 - k_3) = 0]$$

Note: Equation (1) has two other solutions $z=0$ and $z=-1$ also, which are not acceptable as $|z(z+1)| \neq 0$ is given.]

Q.15 (D) 3 2 4 1

$$\text{Sol. } |z-1|=2 \Rightarrow (z-1)(\bar{z}-1)=4 \Rightarrow \frac{4}{\bar{z}-1}=z-1$$

$$z' = \frac{3}{3 - \frac{4}{\bar{z}-1}} \Rightarrow z' = \frac{3}{3 - (z-1)} \Rightarrow 4 - z = \frac{3}{z'} \Rightarrow z = 4 - \frac{3}{z'}$$

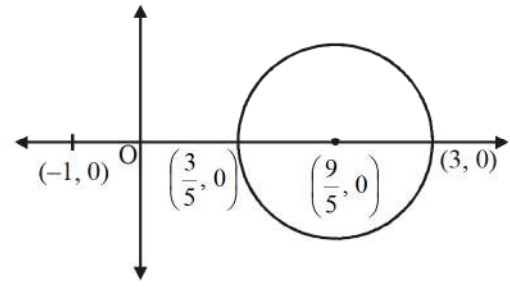
$$\text{Now, } |z-1|=2 \Rightarrow \left| 4 - \frac{3}{z'} - 1 \right| = 2 \Rightarrow 3|z'-1| = 2|z'|$$

Taking $z' = x + iy$

$$9((x-1)^2 + y^2) = 4(x^2 + y^2)$$

$$5(x^2 + y^2) - 18x + 9 = 0$$

$$\therefore x^2 + y^2 - \frac{18}{5}x + \frac{9}{5} = 0$$



$$\text{Centre: } \left(\frac{9}{5}, 0\right), r = \sqrt{\left(\frac{9}{5}\right)^2 - \frac{9}{5}} = \frac{6}{5}$$

$$|z' + 1|_{\max.} = 4$$

$$|z'|_{\min.} = \frac{3}{5} \quad \text{Ans.}$$

Q.16 (C) 1 2 3 4
Sol.

$$(P) \quad \text{We have, } \alpha = \frac{z - \bar{\omega}}{16 + z\bar{\omega}} \Rightarrow \bar{\alpha} = \frac{\bar{z} - \omega}{16 + \bar{z}\omega}$$

$$\text{But } z\bar{z} = \omega\bar{\omega} = 16$$

$$\text{So, } \bar{\alpha} = \frac{\frac{16}{z} - \frac{16}{\bar{\omega}}}{16 + \left(\frac{16}{z}\right)\left(\frac{16}{\bar{\omega}}\right)} = -\alpha \Rightarrow \alpha + \bar{\alpha} = 0 \Rightarrow \text{Re}(\alpha) = 0.$$

$$(Q) \quad x = \frac{x^3}{x^2} = \left(\frac{2+11i}{3+4i}\right) \times \frac{(3-4i)}{(3-4i)} = (2+i)$$

$$\therefore p = 2, q = 1$$

$$\Rightarrow (p+q) = 2+1 = 3 \quad \text{Ans.}$$

$$(R) \quad \text{We have, } \bar{z} = iz^2; \quad \text{Put } z = x + iy$$

$$\Rightarrow x - iy = i(x^2 - y^2 + 2ixy)$$

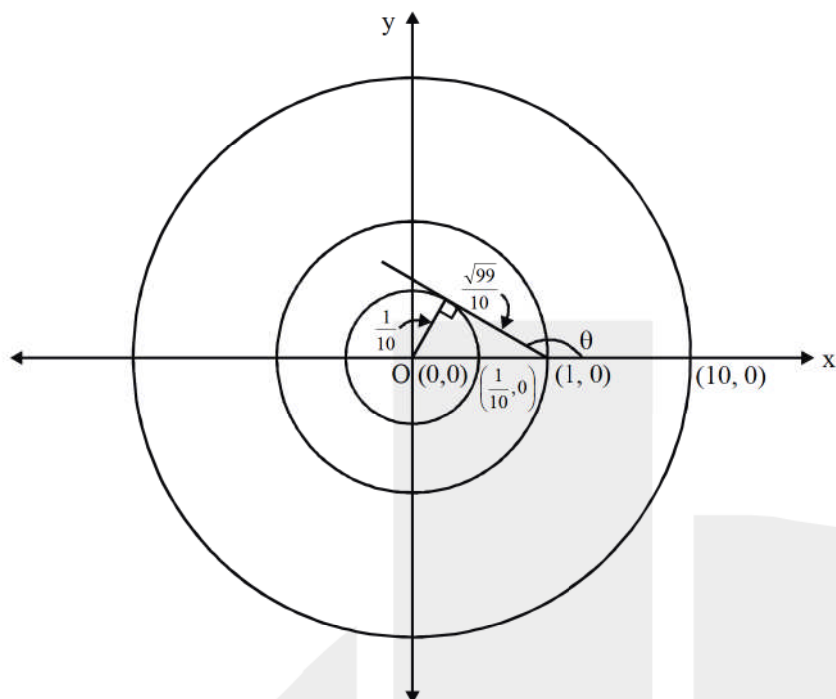
$\therefore$ On solving, we get

$$z = 0 + i0, i, \frac{\sqrt{3}}{2} - \frac{i}{2}, \frac{-\sqrt{3}}{2} - \frac{i}{2}$$

$\Rightarrow$ 4 solutions **Ans.**

$$(S) \quad |3z + 9 - 7i| = |(3z + 6 - 3i) + (3 - 4i)| \\ \leq |3z + 6 - 3i| + |3 - 4i| \\ = 3|z + 2 - i| + 5 \\ = 20$$

Q.17 1
Sol.



Put $\frac{z_1}{z_2} = z$

$$\therefore \left| \frac{z_1}{z_2} \right| = |z| \Rightarrow |z| = \frac{1}{10}$$

Also, $\theta = \arg\left(\frac{z_1}{z_2} - 1\right) = \arg(z - 1)$

$$\therefore \tan^2 \theta \Big|_{\max.} = \frac{1/100}{99/100} = \frac{1}{99} \equiv \frac{m}{n} \text{ (Given).}$$

Q.18 100

Sol. Let $S = 1 + \frac{2}{z} + \frac{3}{z^2} + \frac{4}{z^3} + \dots$

$$\frac{S}{z} = \frac{1}{z} + \frac{2}{z^2} + \frac{3}{z^3} + \dots$$

$$\frac{S}{z} - \frac{S}{z} = \frac{1}{z} + \frac{2}{z^2} + \frac{3}{z^3} + \dots - \left(\frac{1}{z} + \frac{2}{z^2} + \frac{3}{z^3} + \dots \right)$$

$$S \left(1 - \frac{1}{z} \right) = 1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \dots$$

$$\frac{S(z-1)}{z} = \frac{1}{1 - \frac{1}{z}} = \frac{z}{z-1}$$

$$S = \frac{z^2}{(z-1)^2}$$

$$\therefore iz^2 = \frac{z^2}{(z-1)^2} \quad (z \neq 0)$$

$$i = \frac{1}{(z-1)^2} \Rightarrow (z-1)^2 = \frac{1}{i} = -i \Rightarrow z-1 = \pm \sqrt{-i}$$

$$z = 1 \pm \sqrt{-i}$$

$$\text{given } z = n \pm \sqrt{-i}$$

$$\Rightarrow 100n = 100$$

Q.19 4

$$\text{Sol. } z^{2014} \Rightarrow 1$$

$$z = e^{\left(\frac{2k\pi}{2014}\right)i} = \alpha^k \quad (\text{As given in the question})$$

$$\text{Let } z_1 = re^{i\theta_1} \text{ and } z_2 = re^{i\theta_2} \text{ where } r = \frac{1}{\sqrt{1007}}$$

$$\therefore |z_1 + \alpha^k z_2|^2 = (z_1 + \alpha^k z_2)(\bar{z}_1 + \overline{\alpha^k z_2}) = r^2 \left(e^{i\theta_1} + e^{\left(\frac{2k\pi}{2014} + \theta_2\right)i} \right) \left(e^{-i\theta_1} + e^{-i\left(\frac{2k\pi}{2014} + \theta_2\right)} \right)$$

$$= r^2 \left[2 + 2 \cos \left(\theta_1 - \theta_2 - \frac{k\pi}{2007} \right) \right]$$

$$\text{Let } \theta_1 - \theta_2 = \theta \text{ and } \beta = \frac{\pi}{2007}$$

$$\sum_{k=0}^{2013} |z_1 + \alpha^k z_2|^2 = 2r^2 \left[2014 + \underbrace{\cos \theta + \cos(\theta - \alpha) + \cos(\theta - 2\alpha) + \dots + \cos(\theta - 2013\alpha)}_0 \right]$$

$$= 2 \times \frac{1}{1007} \times 2014 = 4$$

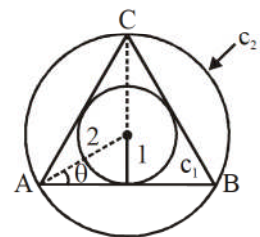
Q.20 6

$$\text{Sol. } \left| \frac{z + \bar{z}}{2\operatorname{Re}(z)} \right| - |z| = \frac{2}{|\bar{z}|} - \left| \frac{z - \bar{z}}{\operatorname{Im}(z)} \right|$$

$$1 = |z| + \frac{2}{|\bar{z}|} - 2$$

$$|z| + \frac{2}{|\bar{z}|} = 3 \Rightarrow |z|^2 - 3|z| + 2 = 0 \Rightarrow |z| = 1 \text{ or } |z| = 2$$

From the figure



$$\sin \theta = \frac{1}{2} \Rightarrow \theta = 30^\circ \Rightarrow \Delta ABC \text{ is equilateral}$$

$$AB = 2\sqrt{3}$$

$$\therefore \text{Ar}(\Delta ABC) = \frac{\sqrt{3}}{4} \cdot 4 \cdot 3 = \sqrt{27} = \sqrt{k}$$

$$\therefore \left[\frac{k}{4} \right] = \left[\frac{27}{4} \right] = 6$$

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